

MH1300 FOUNDATIONS OF MATHEMATICS

Tutorial 11 Solutions

Ex. 7.1.42, 43, 48. Let X and Y be sets, let A and B be any subsets of X and let C and D be any subsets of Y . Determine whether the properties are true for all functions F from X to Y and whether the properties are false for at least one function F from X to Y . Justify your answer.

42. $F(A \cap B) \subseteq F(A) \cap F(B)$.

43. $F(A) \cap F(B) \subseteq F(A \cap B)$.

48. For all subsets C and D of Y ,

$$F^{-1}(C - D) = F^{-1}(C) - F^{-1}(D).$$

SOLUTION . 42. This property is true.

Proof:

Let $F : X \rightarrow Y$ be any function, and suppose $A \subseteq X$ and $B \subseteq X$.
Let $y \in F(A \cap B)$. Then $y = F(x)$ for some $x \in A \cap B$. By definition of intersection, $x \in A$ and $x \in B$. So,

$$y = F(x) \quad \text{for some } x \in A \text{ (in which case } y \in F(A))$$

and

$$y = F(x) \quad \text{for the same } x \in B \text{ (in which case } y \in F(B)).$$

Hence by definition of intersection, $y \in F(A) \cap F(B)$.

43. This property is false.

Counterexample:

Let $X = \{0, 1\}$, $Y = \{0\}$, $A = \{0\}$, $B = \{1\}$, and let $F : X \rightarrow Y$ be defined by $F(0) = F(1) = 0$.

Then $0 \in \{0\} = F(\{0\}) \cap F(\{1\}) = F(A) \cap F(B)$ but $A \cap B = \{0\} \cap \{1\} = \emptyset$. Therefore $0 \notin F(A \cap B) = \emptyset$.

48. The statement is true.

Proof:

Let $f : X \rightarrow Y$ be any function, and suppose that $C \subseteq Y$ and $D \subseteq Y$. For any element $x \in X$, by definition of inverse image and set difference,

$$\begin{aligned} x \in f^{-1}(C - D) &\iff f(x) \in C - D \\ &\iff f(x) \in C \text{ and } f(x) \notin D \\ &\iff x \in f^{-1}(C) \text{ and } x \notin f^{-1}(D) \\ &\iff x \in f^{-1}(C) - f^{-1}(D). \end{aligned}$$

Hence $f^{-1}(C - D) = f^{-1}(C) - f^{-1}(D)$.

Remark: There are several similar problems in Ex. 7.1.41–49. Not all of them are true. □

Ex. 7.2.13.

a. Define $H : \mathbb{R} \rightarrow \mathbb{R}$ by the rule $H(x) = x^2$, for all real numbers x .

(i) Is H one-to-one? Prove or give a counterexample.

- (ii) Is H onto? Prove or give a counterexample.
 b. Define $K : \mathbb{R}^{\text{nonneg}} \rightarrow \mathbb{R}^{\text{nonneg}}$ by the rule $K(x) = x^2$, for all nonnegative real numbers x . Is K onto? Prove or give a counterexample.

SOLUTION . a. (i) No, H is not one-to-one. E.g., take $x_1 = 1$ and $x_2 = -1$. Then $H(1) = 1^2 = 1$ and $H(-1) = (-1)^2 = 1$. Therefore $H(1) = H(-1)$ but $1 \neq -1$.

- (ii) No, H is not onto. Consider $y = -1$ in the co-domain $\mathbb{R}$. Suppose $y = -1$ has a pre-image, say x , then $H(x) = x^2 = -1$. Hence $x^2 = -1$ for some real number x , which is impossible. Hence $y = -1$ has no pre-image, and so H is not onto.

- b. K is onto.

Proof:

Suppose y is any element of $\mathbb{R}^{\text{nonneg}}$. Let $x = \sqrt{y}$. Then x is a real number because $y \geq 0$, and by definition of K ,

$$K(x) = (\sqrt{y})^2 = y.$$

□

Ex. 7.2.17. The function f is defined on a set of real numbers. Determine whether or not f is one-to-one and justify your answer.

$$f(x) = \frac{3x-1}{x}, \text{ for all real numbers } x \neq 0.$$

SOLUTION . Yes, f is one-to-one.

Proof:

Let x_1 and x_2 be any real numbers (other than 0) in the domain of f , and suppose that $f(x_1) = f(x_2)$. [Want to show $x_1 = x_2$.]

By definition of f , $f(x_1) = f(x_2)$ means

$$\frac{3x_1-1}{x_1} = \frac{3x_2-1}{x_2}.$$

Cross-multiplying gives

$$(3x_1-1)x_2 = (3x_2-1)x_1,$$

or equivalently,

$$3x_1x_2 - x_2 = 3x_1x_2 - x_1.$$

subtracting $3x_1x_2$ from both sides and multiplying the resultant equation by -1 , we arrive at $x_1 = x_2$. □

Ex. 7.2.36, 37. If $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are functions, then the function $(f + g) : \mathbb{R} \rightarrow \mathbb{R}$ is defined by the formula $(f + g)(x) = f(x) + g(x)$ for all real numbers x .

36. If f and g are both one-to-one, is $f + g$ also one-to-one? Justify your answer.

37. If f and g are both onto, is $f + g$ also onto? Justify your answer.

SOLUTION . 36. No, $f + g$ need not be one-to-one.

Counterexample:

Take $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x$, and $g : \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = -x$. It is clear that both f and g are one-to-one. Then for any $x \in \mathbb{R}$,

$$(f + g)(x) = f(x) + g(x) = x + (-x) = x - x = 0.$$

Take $x_1 = 1$ and $x_2 = 2$. Clearly $x_1 \neq x_2$ but $(f + g)(1) = 0$ and $(f + g)(2) = 0$ and so $(f + g)(1) = (f + g)(2)$.

37. No, $f + g$ need not be onto.

Counterexample:

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x$ and $g(x) = -x$ just like the counterexample in the previous question.

From previous question, for all $x \in \mathbb{R}$, $(f + g)(x) = 0$. And so $f + g$ is not onto. (In particular, we see that $y = 1 \in \mathbb{R}$ does not have a pre-image. $\square$)

Ex. 7.3.18. If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are functions and $g \circ f$ is one-to-one, must f be one-to-one? Prove or give a counterexample.

SOLUTION . Yes, f must be one-to-one. Suppose $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are functions such that $g \circ f$ is one-to-one.

Let x_1 and x_2 be two elements of X such that $f(x_1) = f(x_2)$. Clearly $f(x_1) = f(x_2) \in Y$, and so

$$g(f(x_1)) = g(f(x_2)).$$

Therefore,

$$g \circ f(x_1) = g \circ f(x_2).$$

Since $g \circ f$ is one-to-one, we must have $x_1 = x_2$. This shows that f is one-to-one. □

Ex. 7.3.19. If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are functions and $g \circ f$ is onto, must g be onto? Prove or give a counterexample.

SOLUTION . Yes, g must be onto. Suppose $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are functions such that $g \circ f$ is onto. Let z be an element in Z . Since $g \circ f$ is onto, there exists $x \in X$ such that

$$g \circ f(x) = g(f(x)) = z.$$

Let $y = f(x)$. Clearly then $y \in Y$ and $g(y) = z$. Therefore, there exists an element y in Y such that $g(y) = z$. This shows that g is onto. □

Ex. 7.3.24. Define $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ by the formulas

$$f(x) = x + 3 \quad \text{and} \quad g(x) = -x \quad \text{for all } x \in \mathbb{R}.$$

Find $g \circ f$, $(g \circ f)^{-1}$, g^{-1} , f^{-1} , and $f^{-1} \circ g^{-1}$, and state how $(g \circ f)^{-1}$ and $f^{-1} \circ g^{-1}$ are related.

SOLUTION . It is clear that f and g are one-to-one functions and both f and g are onto functions. Therefore, their inverses exist. By theorem in lecture, we know that $g \circ f$ is also one-to-one and onto, and so the inverse of $g \circ f$ exists.

- The composition function $g \circ f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by

$$g \circ f(x) = g(x + 3) = -(x + 3) = -x - 3 \quad \text{for all } x \in \mathbb{R}.$$

- Let $x \in \mathbb{R}$. Suppose for some $z \in \mathbb{R}$, we have $z = g(f(x))$. Then $z = -x - 3$, i.e. $x = -z - 3$. Therefore, $(g \circ f)^{-1} : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $(g \circ f)^{-1}(z) = -z - 3$ for all $z \in \mathbb{R}$.
- Let $y \in \mathbb{R}$. Suppose for some $z \in \mathbb{R}$, we have $z = g(y)$. Then $z = -y$. Then $y = -z$. Therefore, $g^{-1} : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $g^{-1}(z) = -z$ for all $z \in \mathbb{R}$.
- Let $x \in \mathbb{R}$. Suppose for some $y \in \mathbb{R}$, we have $y = f(x)$. Then $y = x + 3$. Then $x = y - 3$. Therefore, $f^{-1} : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f^{-1}(y) = y - 3$ for all $y \in \mathbb{R}$.
- The composition function $f^{-1} \circ g^{-1} : \mathbb{R} \rightarrow \mathbb{R}$ is defined by

$$f^{-1}(g^{-1}(z)) = f^{-1}(-z) = -z - 3 \quad \text{for all } z \in \mathbb{R}.$$

By the above and the definition of equality of functions, $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$. □